

ADVANCED JAVA LABSIDDHARTHA ACADEMY OF HIGHER EDUCATION (DEEMED TO BE UNIVERSITY)
(23AI5351) • DEPARTMENT OF AI • BATCH 2025–2029

Week 06: Exception Handling & Custom Exceptions

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1. OBJECTIVE / AIM

To implement robust exception handling using try, catch, finally, throw, throws blocks and create user-defined custom exceptions.

2. CONCEPT & THEORY

Exception handling provides a structured mechanism to intercept and manage runtime anomalies, ensuring that program flow does not terminate abruptly.

3. JAVA SOURCE IMPLEMENTATION

```
// Advanced Java Lab - Experiment 06
// Student: Trivikram | Roll No: 25EU02067
class InvalidAgeException extends Exception {
    public InvalidAgeException(String message) {
        super(message);
    }
}

public class ExceptionHandlingDemo {
    static void validateVoter(int age) throws InvalidAgeException {
        if (age < 18) {
            throw new InvalidAgeException("Age " + age + " is ineligible for registration");
        } else {
            System.out.println("Verification Passed: Registration Successful!");
        }
    }

    public static void main(String[] args) {
        int testAge = 16;
        try {
            System.out.println("Verifying candidate age: " + testAge);
            validateVoter(testAge);
        } catch (InvalidAgeException e) {
            System.err.println("Handled Exception: " + e.getMessage());
        } finally {
            System.out.println("Verification pipeline executed cleanly.");
        }
    }
}
```

```
    }  
  }  
}
```

4. COMPILATION & EXECUTION OUTPUT

```
Verifying candidate age: 16  
Handled Exception: Age 16 is ineligible for registration.  
Verification pipeline executed cleanly.
```

